

Product Requirements Document (PRD)

University School

1. Product Overview

Product Name

University School

Tagline / Hero Statement

Study at a U.S. university while still in high school — and come home every day.

Product Type

U.S.-based **University Access, Academic Support, and Progression Platform**
(not an instructional institution in Brazil)

Core Definition

University School is a U.S.-based platform that enables high school and early college students to **enroll in accredited U.S. university courses**, earn **official college credits**, and receive **AI-powered and human academic support**, while completing their local high school or university studies.

University School **does not provide instructional content**.

All instruction, grading, certification, and credit issuance are performed by **accredited U.S. institutions**.

University School operates as:

- an **access layer**
- an **academic support layer**
- a **guidance and progression layer**
- a **physical center (hub/polo) operating system**
- a **franchise-ready platform**

2. Problem Statement

Market Problems

1. Brazilian students face **duplicated coursework** between high school and university.
 2. Access to U.S. university credits is:
 - expensive,
 - bureaucratic,
 - geographically restrictive.
 3. Brazilian online higher education is massive but lacks **international academic portability**.
 4. Bilingual and international schools lack **structured, credit-bearing U.S. pathways**.
 5. Students who work need **flexible, online, night-compatible models**.
 6. Families want **real university experience**, not just admissions consulting.
-

3. Opportunity & Market Size

Brazil – Higher Education

- ~8 million private higher education students
- ~5 million already enrolled in **online programs**
- 6 major online university groups = ~1.4–1.5M students

K–12 Bilingual / International Schools

- ~1,350 schools
- ~500,000 students

- High English proficiency
- High interest in U.S. university pathways

Core Insight

Even **1–2% conversion** represents tens of thousands of students.

4. Target Segments

Primary Segments

1. Bilingual / International K–12 Schools

- Students aged 14–18
- Goal: U.S. universities
- Value: early credits + stronger applications

2. Private Online Higher Education Students

- Working adults
- Night / asynchronous learners
- Goal: portability, acceleration, employability

3. Large Online University Groups

- Distribution via local learning centers (polos)
- White-label integration

Secondary Segment

- Families relocating to the U.S. via **E-2 Visa**
 - Franchisees operating University School Centers
-

5. Value Proposition

University School enables students to:

- Earn **official U.S. college credits** while in high school
- Replace or fulfill **High School graduation requirements**
- Eliminate duplicated coursework
- Accelerate time-to-degree
- Improve U.S. transfer and admissions outcomes
- Study **inside real U.S. university systems**
- Receive **AI + human academic support**
- Maintain local life, family, and school attendance

It is not about being approved — it is about studying at a Top American University now.

6. Academic Model

6.1 High School Credit Requirements Covered

University School aligns U.S. college General Education courses to satisfy:

- U.S. History (1.0)
 - U.S. Government / Civics (0.5)
 - Economics (0.5)
 - Personal Financial Literacy (0.5)
-

6.2 College General Education Requirements

Students complete official college GE credits:

1. Writing / English (2)
 2. Math / Quantitative Reasoning (1)
 3. Natural Science (1)
 4. Social Science (1)
 5. Humanities / Civic / Ethics (1)
-

6.3 Mandatory U.S. Courses (Delivered by Partners)

Arizona State University (ASU)

- ENG 101 – First-Year Composition I
- ENG 102 – First-Year Composition II
- STP 226 – Elements of Statistics
- POS 110 – American Government and Politics
- BIO 130 – Introduction to Environmental Science
- FIN 294 – Fundamentals of Personal Finance
- HST 110 – United States Since 1865

Brigham Young University (BYU)

- ECON 110 – Economic Principles and Problems

All instructional content, grading, and certification occur **inside ASU/BYU platforms**.

7. University School Platform (Core Product)

7.1 Platform Purpose

University School is the **student-facing and center-facing system** that orchestrates:

- enrollment
 - progression
 - academic support
 - mentoring
 - AI guidance
 - engagement
 - reporting
 - franchise operations
-

7.2 Key Platform Modules

A. Student Dashboard

- Course roadmap
- GE credit tracking
- HS credit equivalency
- Deadlines and milestones
- GPA tracking
- Progress visualization

B. AI Guidance Counseling

- Academic planning
- Credit mapping
- University pathway suggestions
- Career exploration
- Admissions readiness

C. AI Tutor

- Course-aligned tutoring
- Writing feedback
- Rubric alignment (ASU/BYU)
- Concept reinforcement
- On-demand academic help

D. Gamified Engagement Layer

- Progress badges
- Milestones
- Streaks
- Cohort challenges
- Learning labs and simulations

E. Human Support (On Demand)

- Academic mentors
- Graduate student tutors
- College counselors
- Escalation from AI → human

8. Career Pathways Module (Post-GE)

After **3 months of Career Search**, students select **University Pathways**, supported by electives from:

- Harvard Extension School
- Berkeley Extension

- UCLA Extension

Pathway Clusters

Health & Life Sciences

Medicine, Dentistry, Veterinary, Nursing, Psychology, Public Health, Biomedicine

Engineering & Technology

Computer Science, Software Engineering, Data Science, Electrical, Mechanical, Civil, Mechatronics

Business, Economics & Law

Business Administration, Economics, Finance, Marketing, Analytics, Pre-Law

Agrarian & Sustainability

Agronomy, Environmental Studies, Sustainability

University School curates **pathway-aligned electives**, but does not issue credits itself.

9. Physical Centers (University School Centers)

Concept

University School Centers are **academic support hubs**, not teaching institutions.

Preferably located:

- inside Brazilian universities
- or premium learning centers

Weekly Model

Study Hall (Mon / Wed / Fri)

- AI academic support
- 1:1 tutoring (graduate students)
- Advising & mentoring

Campus Access (Tue / Thu)

- Learning labs
- Simulations
- Hands-on projects
- University tool usage

Key Rule

! No instructional content is taught at centers.

10. Legal & Compliance Positioning

- University School **does not operate as an educational provider in Brazil**
- All instruction and certification occur **outside Brazil**
- University School provides:
 - access
 - onboarding
 - support
 - mentoring
 - guidance

This model:

- avoids Brazilian regulatory restrictions
 - allows partnerships with private institutions
 - enables franchise scalability
-

11. Franchise Model (E-2 Ready)

Franchise Concept

University School Centers can be franchised to Brazilian entrepreneurs seeking **E-2 Visa** eligibility.

Franchise Provides

- Platform license
- Academic operating system
- Sales & onboarding SOPs
- AI support stack
- Brand & partnerships
- Training & certification

Operational References

- UniCesumar (EAD polos)
 - Score At The Top (U.S.)
 - Alpha School (mentoring model)
-

12. Operations & SOPs

University School integrates:

- tutor routines
- lead management
- onboarding workflows
- student success monitoring
- CRM & reporting

All operational documents provided are incorporated into:

- Franchise Playbook
 - Center SOPs
 - Staff training modules
-

13. Monetization

Revenue Streams

- Tuition / fee split with U.S. partners
- Platform access fees
- Franchise fees
- Upsells:
 - career counseling
 - application support
 - premium mentoring

Scalability

- Online-first
 - Asset-light
 - High-margin
 - Replicable across countries
-

14. Success Metrics (KPIs)

- Credit completion rate

- GPA performance
 - Retention
 - Time-to-degree reduction
 - Franchise performance
 - Student satisfaction (NPS)
 - Conversion to U.S. universities
-

15. Out of Scope

- Creating instructional content
 - Issuing diplomas
 - Acting as a Brazilian school
 - Competing with partner universities
-

16. Strategic Vision

University School is building the **global operating system for early university access**, starting in Brazil, expanding through franchises, and integrating elite U.S. academic ecosystems.

If you want, next I can:

- turn this into an **Investor Deck**
- extract a **Franchise FDD outline**
- create a **Platform Architecture spec**
- or convert this into a **Legal/Compliance memo**

UNIVERSITY SCHOOL

Franchise Disclosure Document (FDD) – STRUCTURAL OUTLINE

(FTC Rule 16 CFR Part 436 Compliant)

ITEM 1 – The Franchisor and Any Parents, Predecessors, and Affiliates

1.1 Franchisor Identity

- Legal name of franchisor: **University School, Inc.**
- Jurisdiction: United States
- Principal business address
- Nature of business:
 - University access platform
 - Academic support services
 - Franchise licensing for learning centers
- Not an educational institution; not a degree-granting entity

1.2 Affiliates

- University School Platform Entity
- Valley Academy (Immersion Programs – separate product line)
- U.S. University Partners (ASU, BYU, Extensions – **not franchise affiliates**)

1.3 Business Experience

- Years in operation
 - Background in:
 - Online education
 - Learning centers
 - Franchise-based academic models
 - International education pathways
-

ITEM 2 – Business Experience

2.1 Key Executives

For each principal:

- Name
- Position
- Education
- Prior experience (last 5–10 years)
- Role in franchise system

2.2 Advisory & Academic Leadership

- Academic directors
 - University liaisons
 - Platform and AI leadership
 - Operations and franchise support leadership
-

ITEM 3 – Litigation

- Disclosure of:
 - Pending litigation
 - Prior litigation
 - Franchise-related litigation
 - Statement if none exist
-

ITEM 4 – Bankruptcy

- Disclosure for:
 - Franchisor
 - Executives
 - Statement if none exist
-

ITEM 5 – Initial Franchise Fee

5.1 Initial Franchise Fee

- One-time franchise license fee
- Covers:
 - Brand license
 - Platform access
 - Training
 - Initial onboarding
 - Territory setup

5.2 Payment Terms

- Due at signing
 - Non-refundable conditions
 - Possible phased payments (if applicable)
-

ITEM 6 – Other Fees

6.1 Ongoing Fees

- Monthly royalty (percentage or flat)
- Platform usage fee
- AI services fee
- CRM / academic system fee

6.2 Additional Fees

- Marketing fund contribution
- Local marketing spend minimum
- Training for additional staff
- Renewal fees

(Aligned with UniCesumar-style operational cost structures)

ITEM 7 – Estimated Initial Investment

Detailed table including:

- Franchise fee
- Facility setup (small office / learning center)
- Furniture & equipment

- Computers & internet
- Branding & signage
- Initial marketing
- Training & travel
- Working capital (3–6 months)

Designed to remain **E-2 Visa compliant** (active, non-marginal business).

ITEM 8 – Restrictions on Sources of Products and Services

8.1 Mandatory Systems

- University School platform
- CRM
- AI guidance systems
- Academic dashboards

8.2 Approved Vendors

- IT
- Branding
- Marketing
- Learning lab equipment

8.3 Academic Content

- No proprietary teaching materials required
- Instruction delivered only by U.S. universities

ITEM 9 – Franchisee’s Obligations

Detailed table covering:

- Site selection
- Staffing
- Marketing
- Student onboarding
- Academic support delivery
- Data reporting
- Compliance with academic integrity standards
- Use of AI + human support protocols

ITEM 10 – Financing

- Whether franchisor offers financing
- Third-party financing references
- E-2 Visa investment guidance (informational, not legal advice)

ITEM 11 – Franchisor’s Assistance, Advertising, Computer Systems, and Training

11.1 Pre-Opening Assistance

- Location guidance
- Center layout standards

- Technology setup
- Hiring guidelines
- Marketing launch kit

11.2 Training Program

- Initial franchisee training
- Academic operations training
- Sales & enrollment training
- CRM & AI system training
- Ongoing training

(Incorporates UniCesumar tutor & polo SOPs)

11.3 Ongoing Support

- Academic escalation
- AI support maintenance
- Marketing updates
- Regulatory guidance updates

ITEM 12 – Territory

- Exclusive or protected territory
 - Territory definition (city / metro)
 - Conditions for expansion
 - Online activity boundaries
-

ITEM 13 – Trademarks

- University School trademark status
 - Logos
 - Branding rules
 - Usage restrictions
-

ITEM 14 – Patents, Copyrights, and Proprietary Information

- Platform IP
 - AI models
 - Academic mapping frameworks
 - SOPs
 - Data structures
-

ITEM 15 – Obligation to Participate in the Actual Operation of the Franchise Business

- Owner-operator expectations
 - Visa-related operational requirements
 - Minimum involvement standards
-

ITEM 16 – Restrictions on What the Franchisee May Sell

- Only approved University School services

- No independent instructional offerings
 - No diploma issuance
 - No unapproved partnerships
-

ITEM 17 – Renewal, Termination, Transfer, and Dispute Resolution

- Franchise term
 - Renewal conditions
 - Termination rights
 - Transfer approval
 - Non-compete clauses
 - Arbitration / jurisdiction
-

ITEM 18 – Public Figures

- Disclosure if none used
-

ITEM 19 – Financial Performance Representations (Optional)

- Optional earnings claims
- Typical center performance ranges
- Disclosure methodology
- Clear disclaimers

ITEM 20 – Outlets and Franchisee Information

- Number of franchises
- Open / closed / transferred units
- Pipeline projections
- Contact info for current franchisees

ITEM 21 – Financial Statements

- Audited financials (3 years or as required)
- Balance sheet
- Income statement
- Cash flow

ITEM 22 – Contracts

List of exhibits:

- Franchise Agreement
 - Platform License Agreement
 - Operations Manual (summary)
 - IP License
 - Marketing Agreement
 - Data Processing Agreement
-

ITEM 23 – Receipts

- Franchisee acknowledgment
 - Signature pages
-

Key Strategic Notes (Non-FDD, Internal)

- Structure fully supports **E-2 Visa**
- Avoids Brazilian educational regulation exposure
- Replicable, asset-light
- AI-first with human escalation
- Strong precedent via UniCesumar EAD polos
- High-margin franchise model

UNIVERSITY SCHOOL

Franchise Disclosure Document (FDD) – STRUCTURAL OUTLINE

(FTC Rule 16 CFR Part 436 Compliant)

ITEM 1 – The Franchisor and Any Parents, Predecessors, and Affiliates

1.1 Franchisor Identity

- Legal name of franchisor: **University School, Inc.**

- Jurisdiction: United States
- Principal business address
- Nature of business:
 - University access platform
 - Academic support services
 - Franchise licensing for learning centers
- Not an educational institution; not a degree-granting entity

1.2 Affiliates

- University School Platform Entity
- Valley Academy (Immersion Programs – separate product line)
- U.S. University Partners (ASU, BYU, Extensions – **not franchise affiliates**)

1.3 Business Experience

- Years in operation
- Background in:
 - Online education
 - Learning centers
 - Franchise-based academic models
 - International education pathways

ITEM 2 – Business Experience

2.1 Key Executives

For each principal:

- Name
- Position
- Education
- Prior experience (last 5–10 years)
- Role in franchise system

2.2 Advisory & Academic Leadership

- Academic directors
 - University liaisons
 - Platform and AI leadership
 - Operations and franchise support leadership
-

ITEM 3 – Litigation

- Disclosure of:
 - Pending litigation
 - Prior litigation
 - Franchise-related litigation
 - Statement if none exist
-

ITEM 4 – Bankruptcy

- Disclosure for:
 - Franchisor
 - Executives

- Statement if none exist
-

ITEM 5 – Initial Franchise Fee

5.1 Initial Franchise Fee

- One-time franchise license fee
- Covers:
 - Brand license
 - Platform access
 - Training
 - Initial onboarding
 - Territory setup

5.2 Payment Terms

- Due at signing
 - Non-refundable conditions
 - Possible phased payments (if applicable)
-

ITEM 6 – Other Fees

6.1 Ongoing Fees

- Monthly royalty (percentage or flat)
- Platform usage fee
- AI services fee

- CRM / academic system fee

6.2 Additional Fees

- Marketing fund contribution
- Local marketing spend minimum
- Training for additional staff
- Renewal fees

(Aligned with UniCesumar-style operational cost structures)

ITEM 7 – Estimated Initial Investment

Detailed table including:

- Franchise fee
- Facility setup (small office / learning center)
- Furniture & equipment
- Computers & internet
- Branding & signage
- Initial marketing
- Training & travel
- Working capital (3–6 months)

Designed to remain **E-2 Visa compliant** (active, non-marginal business).

ITEM 8 – Restrictions on Sources of Products and Services

8.1 Mandatory Systems

- University School platform
- CRM
- AI guidance systems
- Academic dashboards

8.2 Approved Vendors

- IT
- Branding
- Marketing
- Learning lab equipment

8.3 Academic Content

- No proprietary teaching materials required
- Instruction delivered only by U.S. universities

ITEM 9 – Franchisee’s Obligations

Detailed table covering:

- Site selection
- Staffing
- Marketing
- Student onboarding
- Academic support delivery
- Data reporting

- Compliance with academic integrity standards
 - Use of AI + human support protocols
-

ITEM 10 – Financing

- Whether franchisor offers financing
 - Third-party financing references
 - E-2 Visa investment guidance (informational, not legal advice)
-

ITEM 11 – Franchisor’s Assistance, Advertising, Computer Systems, and Training

11.1 Pre-Opening Assistance

- Location guidance
- Center layout standards
- Technology setup
- Hiring guidelines
- Marketing launch kit

11.2 Training Program

- Initial franchisee training
- Academic operations training
- Sales & enrollment training
- CRM & AI system training
- Ongoing training

(Incorporates UniCesumar tutor & polo SOPs)

11.3 Ongoing Support

- Academic escalation
 - AI support maintenance
 - Marketing updates
 - Regulatory guidance updates
-

ITEM 12 – Territory

- Exclusive or protected territory
 - Territory definition (city / metro)
 - Conditions for expansion
 - Online activity boundaries
-

ITEM 13 – Trademarks

- University School trademark status
 - Logos
 - Branding rules
 - Usage restrictions
-

ITEM 14 – Patents, Copyrights, and Proprietary Information

- Platform IP

- AI models
 - Academic mapping frameworks
 - SOPs
 - Data structures
-

ITEM 15 – Obligation to Participate in the Actual Operation of the Franchise Business

- Owner-operator expectations
 - Visa-related operational requirements
 - Minimum involvement standards
-

ITEM 16 – Restrictions on What the Franchisee May Sell

- Only approved University School services
 - No independent instructional offerings
 - No diploma issuance
 - No unapproved partnerships
-

ITEM 17 – Renewal, Termination, Transfer, and Dispute Resolution

- Franchise term
- Renewal conditions
- Termination rights

- Transfer approval
 - Non-compete clauses
 - Arbitration / jurisdiction
-

ITEM 18 – Public Figures

- Disclosure if none used
-

ITEM 19 – Financial Performance Representations (Optional)

- Optional earnings claims
 - Typical center performance ranges
 - Disclosure methodology
 - Clear disclaimers
-

ITEM 20 – Outlets and Franchisee Information

- Number of franchises
 - Open / closed / transferred units
 - Pipeline projections
 - Contact info for current franchisees
-

ITEM 21 – Financial Statements

- Audited financials (3 years or as required)
 - Balance sheet
 - Income statement
 - Cash flow
-

ITEM 22 – Contracts

List of exhibits:

- Franchise Agreement
 - Platform License Agreement
 - Operations Manual (summary)
 - IP License
 - Marketing Agreement
 - Data Processing Agreement
-

ITEM 23 – Receipts

- Franchisee acknowledgment
 - Signature pages
-

Key Strategic Notes (Non-FDD, Internal)

- Structure fully supports **E-2 Visa**
- Avoids Brazilian educational regulation exposure

- Replicable, asset-light
- AI-first with human escalation
- Strong precedent via UniCesumar EAD polos
- High-margin franchise model

CONSOLIDATED PRODUCT REQUIREMENTS DOCUMENT (PRD)

University School

1. Product Overview

Product Name

University School

Hero Statement

Study at a U.S. university while still in high school — and come home every day.

Product Type

U.S.-based **University Access, Academic Support, and Progression Platform**
(Non-instructional. Not an educational institution in Brazil.)

Core Definition

University School is a **U.S.-based academic access and operating platform** that enables high school and early college students to:

- enroll in **accredited U.S. university courses**,
- earn **official college credits**, and
- receive **AI-powered and human academic support**,

while completing their local high school or university studies in their home country.

University School does not provide instructional content.

All instruction, grading, certification, and credit issuance are delivered **exclusively** by accredited U.S. partner institutions.

University School Operates As:

- an **access layer** (to U.S. universities),
 - an **academic support layer** (AI + human),
 - a **guidance and progression layer** (career + pathways),
 - a **physical center operating system** (polos / hubs),
 - a **franchise-ready platform** (E-2 compliant).
-

2. Problem Statement

Core Market Problems

- 1. Duplicated coursework**
 - Students repeat content between high school and university.
 - Time and money are wasted.
- 2. Barriers to U.S. university credits**
 - High cost
 - Bureaucracy
 - Physical relocation requirements
- 3. Lack of academic portability**
 - Brazilian online higher education is massive but largely non-transferable internationally.
- 4. Gap in bilingual / international schools**

- Strong English proficiency
- No structured, credit-bearing U.S. pathways

5. Adult learners need flexibility

- Night-compatible
- Online-first
- Career-aligned

6. Families want real university experience

- Not just admissions consulting
- Actual academic engagement and credits

3. Opportunity & Market Size

Brazil – Higher Education

- ~8M private higher-education students
- ~5M already enrolled in online programs
- 6 major online groups = ~1.4–1.5M students

K–12 Bilingual / International Schools

- ~1,350 schools
- ~500,000 students
- High English proficiency
- High intent toward U.S. universities

Core Insight

Even **1–2% penetration** represents **tens of thousands of students** with strong LTV.

4. Target Segments

Primary Segments

1. Bilingual / International K–12 Students (14–18)

- Goal: U.S. universities
- Value: early credits, acceleration, stronger applications

2. Private Online Higher Education Students

- Working adults
- Online / night learners
- Value: portability, speed, employability

3. Large Online University Groups

- Distribution via existing local learning centers (polos)
- White-label or co-branded integration

Secondary Segment

- Entrepreneurs seeking **E-2 Visa** via University School franchises
- Families relocating internationally

5. Value Proposition

University School enables students to:

- Earn **official U.S. college credits** early
- Fulfill or replace **High School graduation requirements**

- Eliminate duplicated coursework
- Accelerate **time-to-degree**
- Improve U.S. transfer and admissions outcomes
- Study **inside real U.S. university systems**
- Receive **AI + human academic support**
- Maintain local life, school, and family

It is not about being approved.

It is about studying at a Top American University now.

6. Academic Model

6.1 High School Credits Covered

University School aligns U.S. college General Education (GE) courses to satisfy:

- U.S. History (1.0)
 - U.S. Government / Civics (0.5)
 - Economics (0.5)
 - Personal Financial Literacy (0.5)
-

6.2 College General Education Requirements

Students complete **official college GE credits**:

- Writing / English (2)
- Math / Quantitative Reasoning (1)
- Natural Science (1)
- Social Science (1)

- Humanities / Civic / Ethics (1)
-

6.3 Mandatory U.S. Courses (Partner-Delivered)

Arizona State University (ASU)

- ENG 101 – First-Year Composition I
- ENG 102 – First-Year Composition II
- STP 226 – Elements of Statistics
- POS 110 – American Government and Politics
- BIO 130 – Introduction to Environmental Science
- FIN 294 – Fundamentals of Personal Finance
- HST 110 – United States Since 1865

Brigham Young University (BYU)

- ECON 110 – Economic Principles and Problems

All instruction, grading, and certification occur inside partner university platforms.

7. University School Platform (Core Product)

7.1 Platform Purpose

The platform orchestrates:

- enrollment,
- academic progression,
- support delivery,
- AI guidance,

- engagement,
 - reporting,
 - franchise operations.
-

7.2 Key Platform Modules

A. Student Dashboard

- Course roadmap
- GE credit tracking
- HS equivalency mapping
- Deadlines & milestones
- GPA tracking
- Progress visualization

B. AI Guidance Counseling

- Academic planning
- Credit mapping
- University pathway recommendations
- Career exploration
- Admissions readiness

C. AI Tutor

- Course-aligned tutoring
- Writing feedback
- Rubric alignment (ASU/BYU)
- Concept reinforcement

- On-demand academic help

D. Gamified Engagement Layer

- Badges & milestones
- Progress streaks
- Cohort challenges
- Learning labs & simulations

E. Human Support (On-Demand)

- Academic mentors
 - Graduate student tutors
 - College counselors
 - AI → human escalation
-

8. Career Pathways Module (Post-GE)

After a **3-month Career Search phase**, students select **University Pathways**.

Elective Providers

- Harvard Extension School
- Berkeley Extension
- UCLA Extension

Pathway Clusters

Health & Life Sciences

Medicine, Dentistry, Veterinary, Nursing, Psychology, Public Health, Biomedicine

Engineering & Technology

CS, Software Engineering, Data Science, Electrical, Mechanical, Civil, Mechatronics

Business, Economics & Law

Business, Economics, Finance, Marketing, Analytics, Pre-Law

Agrarian & Sustainability

Agronomy, Environmental Studies, Sustainability

University School **curates and guides**, but does not issue credits.

9. Physical Centers (University School Centers)

Concept

Centers are **academic support hubs**, not teaching institutions.

Preferred Locations

- Inside Brazilian universities
- Premium learning centers
- Professional office environments

Weekly Model

Study Hall (Mon / Wed / Fri)

- AI academic support
- 1:1 tutoring
- Advising & mentoring

Campus Access (Tue / Thu)

- Learning labs
- Simulations
- Hands-on projects
- University tools

! No instructional content is taught at centers.

10. Legal & Compliance Positioning

- University School is **not an educational provider in Brazil**
- All instruction and certification occur **outside Brazil**
- University School provides:
 - access
 - onboarding
 - support
 - mentoring
 - guidance

This structure:

- avoids Brazilian regulatory exposure
- enables partnerships
- supports franchising and scale

11. Franchise Model (E-2 Ready)

Franchise Concept

University School Centers may be franchised to entrepreneurs seeking **E-2 Visa eligibility**.

Franchise Provides

- Platform license
- Academic operating system
- Sales & onboarding SOPs

- AI support stack
- Brand & partnerships
- Training & certification

Operational References

- UniCesumar (EAD polos)
 - Score At The Top (U.S.)
 - Alpha School (mentoring)
-

12. Operations & SOPs

Integrated from all provided documents:

- tutor routines
- lead management
- onboarding workflows
- student success monitoring
- CRM & reporting

Delivered as:

- Franchise Playbook
 - Center SOPs
 - Staff training modules
-

13. Monetization

Revenue Streams

- Tuition / fee split with U.S. partners
- Platform access fees
- Franchise fees
- Upsells:
 - career counseling
 - application support
 - premium mentoring

Scalability

- Online-first
 - Asset-light
 - High-margin
 - Globally replicable
-

14. Success Metrics (KPIs)

- Credit completion rate
 - GPA performance
 - Retention
 - Time-to-degree reduction
 - Franchise performance
 - Student NPS
 - Conversion to U.S. universities
-

15. Out of Scope

- Creating instructional content
 - Issuing diplomas
 - Acting as a Brazilian school
 - Competing with partner universities
-

16. Strategic Vision

University School is building the global operating system for early university access, starting in Brazil, scaling through franchises, and integrating elite U.S. academic ecosystems.

UNIVERSITY SCHOOL

Platform Architecture Specification & Engineering PRD

(Tech-Only)

1. System Overview

1.1 System Purpose

The University School platform is a **multi-tenant academic orchestration system** that:

- connects students to **external accredited university LMSs** (ASU, BYU, Extensions),
- tracks academic progress and credit equivalencies,
- delivers **AI-driven guidance, tutoring, and engagement**,

- supports **human escalation**,
- operates **physical centers (polos)**,
- scales via **franchise architecture**.

! The platform **does not host instructional content** or grading systems.

1.2 High-Level Architecture

Architecture Style

- Modular, API-first
- Cloud-native
- Event-driven where applicable
- Multi-tenant (Franchise → Center → Student)

Core Layers

1. Presentation Layer (Web / Mobile)
 2. Application Layer (Business Logic)
 3. AI Services Layer
 4. Integration Layer (Universities, CRM, Payments)
 5. Data Layer
 6. Security & Compliance Layer
 7. DevOps & Observability
-

2. User Roles & Permissions

2.1 User Roles

Role	Description
Student	Learner enrolled via University School
Parent/Guardian	View-only / financial permissions
Academic Mentor	Human tutor / advisor
Guidance Counselor	Admissions & pathway advisor
Center Staff	On-site support
Franchise Owner	Business + operational access
Platform Admin	Global admin
AI Agent	System-level service account

2.2 Permission Model

- Role-Based Access Control (RBAC)
 - Attribute-Based Access Control (ABAC) for:
 - geography
 - franchise
 - age
 - data sensitivity
-

3. Core Platform Modules (Engineering Scope)

3.1 Identity & Access Management (IAM)

Requirements

- SSO (OAuth 2.0 / OpenID Connect)
- MFA (role-based enforcement)
- External LMS SSO passthrough (ASU/BYU)
- Franchise-level identity partitioning

Suggested Stack

- Auth0 / Cognito / Azure AD B2C
 - JWT tokens
 - Session isolation per tenant
-

3.2 Student Information System (SIS – Lightweight)

Purpose

Academic orchestration, not instruction.

Core Entities

- Student
- Enrollment
- Course (external reference)
- Credit Unit
- Pathway
- Milestone
- Support Interaction

Key Functions

- Enrollment tracking
- Credit equivalency mapping

- HS ↔ College alignment
 - Progress status
 - GPA ingestion (read-only)
-

3.3 External University Integration Layer

Universities

- ASU
- BYU
- Harvard Extension
- Berkeley Extension
- UCLA Extension

Integration Pattern

- LTI where available
- Secure deep links
- Progress polling or webhook ingestion
- Manual override for institutions without APIs

Data Ingested

- Enrollment confirmation
- Assignment completion
- Final grades
- Credit completion

! No content ingestion or storage.

3.4 Academic Mapping Engine

Purpose

Translate external university credits into:

- HS requirements
- College GE requirements
- Pathway readiness

Logic

- Rule-based engine (v1)
- ML-assisted recommendations (v2)

Outputs

- “HS Requirement Fulfilled”
 - “College GE Completed”
 - “Pathway Eligible”
-

3.5 AI Guidance Counseling Engine

Functions

- Academic planning
- Credit sequencing
- Pathway recommendation
- Risk detection (dropout, overload)

Models

- LLM (multi-provider capable)
- Vector DB for:
 - student profiles
 - pathways
 - policies

Guardrails

- No academic instruction
 - No grading
 - Escalation triggers
-

3.6 AI Tutor (Non-Instructional)

Capabilities

- Writing feedback (rubric-aligned)
- Concept clarification
- Study planning
- Assignment understanding

Constraints

- No original answers submission
 - No plagiarism
 - Academic integrity filters
-

3.7 Gamification & Engagement Engine

Mechanics

- Badges
- Streaks
- Milestones
- Cohort challenges

Triggers

- Course progress
- Attendance (center check-in)
- Support interactions

3.8 Human Support & Escalation System

Flow

AI detects:

- low progress
- repeated confusion
- deadline risk

→ Ticket created

→ Assigned to mentor/counselor

→ SLA-based resolution

Tools

- Internal ticketing
 - Chat + async messaging
 - Session logs
-

3.9 Career Pathway Engine

Functions

- Career interest profiling
- Pathway clustering
- Elective recommendations
- University alignment

Inputs

- Student behavior
 - Assessments
 - Market demand data
-

3.10 Franchise & Center Management System

Capabilities

- Multi-tenant hierarchy
- Center performance dashboards
- Staff permissions
- Revenue reporting
- Compliance tracking

Metrics

- Enrollment
- Retention
- Completion

- SLA adherence
-

4. Data Architecture

4.1 Data Types

Type	Storage
PII	Encrypted relational DB
Academic metadata	Relational DB
Events	Event store / Kafka
AI embeddings	Vector DB
Logs	Centralized logging

4.2 Data Privacy & Compliance

- GDPR
 - LGPD
 - FERPA alignment
 - Data residency controls
 - Consent management
-

5. Non-Functional Requirements

5.1 Scalability

- Horizontal scaling

- Franchise-ready
- Global CDN

5.2 Availability

- 99.9% SLA
- Multi-region support

5.3 Security

- Zero-trust architecture
- Encryption at rest & in transit
- Audit logs

5.4 Observability

- Metrics
- Tracing
- Alerting

6. DevOps & Infrastructure

Cloud

- AWS / GCP / Azure (agnostic)

CI/CD

- Automated testing
- Canary deployments

- Feature flags

Infra as Code

- Terraform / Pulumi
-

7. Engineering Roadmap

Phase 1 – MVP (0–6 months)

- Core platform
- ASU/BYU integration
- Student dashboard
- AI guidance v1
- Franchise base

Phase 2 – Scale (6–12 months)

- Pathways engine
- Advanced AI tutor
- Multi-university expansion
- White-label support

Phase 3 – Optimization (12–24 months)

- Predictive analytics
- Adaptive pathways
- Global rollout

8. Out of Scope (Strict)

- Hosting classes
- Teaching content
- Issuing grades or diplomas
- Competing LMS functionality

9. Key Engineering Principles

- AI-first, human-verified
- Compliance-by-design
- Partner-respectful (universities)
- Franchise-scalable
- Asset-light

Final Note

This platform is **not an LMS**.
It is a **University Access Operating System**.

1) System architecture diagrams (C4)

C4 — Level 1: System Context

flowchart LR

Student[Student] -->|Uses| US[University School Platform]

Parent[Parent/Guardian] -->|Views/Payments| US

Mentor[Human Mentor/Counselor] -->|Supports via| US

CenterStaff[Center Staff] -->|Operates| US

Franchisee[Franchise Owner] -->|Runs business via| US

US -->|Deep links / SSO / progress ingest| ASU[ASU / Partner University LMS]

US -->|Deep links / SSO / progress ingest| BYU[BYU LMS]

US -->|Deep links / course catalog| Ext[Extensions: Harvard/Berkeley/UCLA]

US -->|Payments| PSP[Payment Processor]

US -->|Email/SMS/WhatsApp| MSG[Messaging Provider]

US -->|Analytics| ANA[Analytics/Warehouse]

US -->|Identity/SSO| IAM[Identity Provider]

C4 — Level 2: Container Diagram

flowchart TB

subgraph Client["Client Apps"]

Web[Web App]

Mobile[Mobile App]

AdminPortal[Admin/Franchise Portal]

end

subgraph Platform["University School Platform"]

APIGW[API Gateway]

Auth[Auth Service (OIDC)]

SIS[SIS Service (Student + Enrollment + Progress)]

Mapping[Credit Mapping Engine]

Pathways[Career/Pathways Service]

Gamify[Gamification Service]

```
    Support[Support/Ticketing Service]
    Integrations[Integrations Service]
    Notifications[Notifications Service]
    AIOrch[AI Orchestrator]
    Audit[Audit/Compliance Logging]
end
```

```
subgraph Data["Data Layer"]
    RDB[(Relational DB)]
    Cache[(Redis/Cache)]
    Events[(Event Bus)]
    Vector[(Vector DB)]
    Obj[(Object Storage)]
    DW[(Warehouse/BI)]
end
```

```
subgraph External["External Systems"]
    IdP[Identity Provider]
    ASU[ASU LMS]
    BYU[BYU LMS]
    Ext[Extensions Providers]
    PSP[Payments]
    MSG[Email/SMS/WhatsApp]
end
```

```
Web --> APIGW
Mobile --> APIGW
AdminPortal --> APIGW
```

```
APIGW --> Auth
APIGW --> SIS
APIGW --> Mapping
APIGW --> Pathways
APIGW --> Gamify
APIGW --> Support
APIGW --> Integrations
APIGW --> Notifications
APIGW --> AIOrch
APIGW --> Audit
```

```
SIS --> RDB
```

Mapping --> RDB
Pathways --> RDB
Gamify --> RDB
Support --> RDB
Auth --> RDB
AIOrch --> Vector
AIOrch --> Obj
Integrations --> Events
SIS --> Events
Events --> DW
Notifications --> MSG
Integrations --> ASU
Integrations --> BYU
Integrations --> Ext
Auth --> IdP
SIS --> Cache

C4 — Level 3: Component Diagram (AI Orchestrator)

flowchart LR

```
subgraph AIOrch["AI Orchestrator"]
    Policy[Policy & Guardrails]
    Router[Model Router]
    RAG[RAG Retrieval Layer]
    Prompt[Prompt Builder]
    Memory[Student Context Store]
    Eval[Safety + Quality Eval]
    Log[AI Audit Logger]
end
```

Router --> Prompt
Prompt --> Policy
Policy --> Eval
Prompt --> RAG
RAG --> Vector[(Vector DB)]
RAG --> Obj[(Object Storage)]
Memory --> RDB[(Relational DB)]
Eval --> Router
Eval --> Log
Log --> Audit[(Audit Store)]

2) API Contract Spec (OpenAPI-style)

Below is a **minimal-but-complete** contract for MVP + franchise scaling. (You can expand it directly into a full OpenAPI file.)

Core API principles

- **REST + JSON** (v1), optionally GraphQL later for dashboards.
- **Idempotency keys** for payments & enrollment operations.
- **Tenant scoping** on every request (franchise_id, center_id).
- **PII minimized**; strict RBAC.

Base

- Base URL: `/api/v1`
- Auth: `Authorization: Bearer <JWT>`
- Tenant headers:
 - `X-Franchise-Id: <uuid>` (required for franchise/center staff)
 - `X-Center-Id: <uuid>` (required for center operations)

OpenAPI excerpt

```
openapi: 3.0.3
info:
  title: University School Platform API
  version: "1.0"
servers:
  - url: /api/v1

paths:
  /auth/token:
```

```
post:
  summary: Exchange authorization code for JWT
  requestBody:
    required: true
    content:
      application/json:
        schema:
          type: object
          required: [provider, code, redirect_uri]
          properties:
            provider: { type: string, enum: [oidc] }
            code: { type: string }
            redirect_uri: { type: string }
  responses:
    "200":
      description: JWT issued
      content:
        application/json:
          schema:
            type: object
            properties:
              access_token: { type: string }
              refresh_token: { type: string }
              expires_in: { type: integer }
```

/students:

```
post:
  summary: Create student (admin/center staff)
  security: [{ bearerAuth: [] }]
  requestBody:
    required: true
    content:
      application/json:
        schema:
          $ref: "#/components/schemas/StudentCreate"
  responses:
    "201":
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Student" }
```

```

get:
  summary: List students (scoped by tenant)
  security: [{ bearerAuth: [] }]
  parameters:
    - in: query
      name: q
      schema: { type: string }
    - in: query
      name: center_id
      schema: { type: string, format: uuid }
  responses:
    "200":
      content:
        application/json:
          schema:
            type: object
            properties:
              items:
                type: array
                items: { $ref: "#/components/schemas/Student" }

/students/{student_id}/plan:
get:
  summary: Get academic plan & progress
  security: [{ bearerAuth: [] }]
  parameters:
    - in: path
      name: student_id
      required: true
      schema: { type: string, format: uuid }
  responses:
    "200":
      content:
        application/json:
          schema: { $ref: "#/components/schemas/AcademicPlan" }

/enrollments:
post:
  summary: Create enrollment to external course (records only;
instruction elsewhere)
  security: [{ bearerAuth: [] }]

```

```

    headers:
      Idempotency-Key:
        schema: { type: string }
    requestBody:
      required: true
      content:
        application/json:
          schema: { $ref: "#/components/schemas/EnrollmentCreate" }
  }

  responses:
    "201":
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Enrollment" }

/integrations/lms/events:
  post:
    summary: Ingest LMS progress events (webhook)
    security: [{ bearerAuth: [] }]
    requestBody:
      required: true
      content:
        application/json:
          schema: { $ref: "#/components/schemas/LmsEvent" }
    responses:
      "202": { description: Accepted }

/ai/sessions:
  post:
    summary: Start an AI support session
    security: [{ bearerAuth: [] }]
    requestBody:
      required: true
      content:
        application/json:
          schema: { $ref: "#/components/schemas/AiSessionCreate" }
    responses:
      "201":
        content:
          application/json:
            schema: { $ref: "#/components/schemas/AiSession" }

```

```
/support/tickets:
  post:
    summary: Create support ticket (AI escalation or human request)
    security: [{ bearerAuth: [] }]
    requestBody:
      required: true
      content:
        application/json:
          schema: { $ref: "#/components/schemas/TicketCreate" }
    responses:
      "201":
        content:
          application/json:
            schema: { $ref: "#/components/schemas/Ticket" }

components:
  securitySchemes:
    bearerAuth:
      type: http
      scheme: bearer

  schemas:
    StudentCreate:
      type: object
      required: [first_name, last_name, birthdate, country, preferred_language]
      properties:
        first_name: { type: string }
        last_name: { type: string }
        birthdate: { type: string, format: date }
        country: { type: string }
        preferred_language: { type: string, enum: [en, pt-BR] }
        guardian_email: { type: string }
        center_id: { type: string, format: uuid }

    Student:
      type: object
      properties:
        id: { type: string, format: uuid }
```

```
first_name: { type: string }
last_name: { type: string }
preferred_language: { type: string }
center_id: { type: string, format: uuid }
created_at: { type: string, format: date-time }
```

EnrollmentCreate:

```
type: object
required: [student_id, provider, provider_course_id, term]
properties:
  student_id: { type: string, format: uuid }
  provider: { type: string, enum: [ASU, BYU, HARVARD_EXT,
BERKELEY_EXT, UCLA_EXT] }
  provider_course_id: { type: string }
  term: { type: string }
  deep_link_url: { type: string }
```

Enrollment:

```
type: object
properties:
  id: { type: string, format: uuid }
  student_id: { type: string, format: uuid }
  provider: { type: string }
  provider_course_id: { type: string }
  status: { type: string, enum: [pending, active, completed,
withdrawn] }
```

LmsEvent:

```
type: object
required: [provider, student_id, provider_course_id,
event_type, event_ts]
properties:
  provider: { type: string }
  student_id: { type: string, format: uuid }
  provider_course_id: { type: string }
  event_type: { type: string, enum: [enrolled,
assignment_submitted, grade_posted, completed] }
  payload: { type: object }
  event_ts: { type: string, format: date-time }
```

AcademicPlan:

```
type: object
properties:
  student_id: { type: string, format: uuid }
  hs_requirements:
    type: array
    items: { $ref: "#/components/schemas/RequirementStatus" }
  ge_requirements:
    type: array
    items: { $ref: "#/components/schemas/RequirementStatus" }
  enrollments:
    type: array
    items: { $ref: "#/components/schemas/Enrollment" }
```

```
RequirementStatus:
  type: object
  properties:
    code: { type: string }
    title: { type: string }
    status: { type: string, enum: [not_started, in_progress,
satisfied] }
    satisfied_by_enrollment_id: { type: string, format: uuid }
```

```
AiSessionCreate:
  type: object
  required: [student_id, type]
  properties:
    student_id: { type: string, format: uuid }
    type: { type: string, enum: [tutor, guidance,
writing_feedback] }
    context: { type: object }
```

```
AiSession:
  type: object
  properties:
    id: { type: string, format: uuid }
    student_id: { type: string, format: uuid }
    type: { type: string }
    created_at: { type: string, format: date-time }
```

```
TicketCreate:
  type: object
```

```

    required: [student_id, category, priority, summary]
    properties:
      student_id: { type: string, format: uuid }
      category: { type: string, enum: [academic_support,
counseling, billing, technical, compliance] }
      priority: { type: string, enum: [low, medium, high, urgent]
}
      summary: { type: string }
      details: { type: string }

```

```

Ticket:
  type: object
  properties:
    id: { type: string, format: uuid }
    status: { type: string, enum: [open, assigned, resolved,
closed] }

```

3) Data model ERD (Mermaid)

This ERD supports **multi-tenant franchising**, **non-instructional orchestration**, **AI auditability**, and **support escalation**.

erDiagram

```

FRANCHISE ||--o{ CENTER : has
CENTER ||--o{ STAFF : employs
CENTER ||--o{ STUDENT : serves
STUDENT ||--o{ ENROLLMENT : has
STUDENT ||--o{ AI_SESSION : starts
STUDENT ||--o{ SUPPORT_TICKET : creates
STUDENT ||--o{ STUDENT_ASSESSMENT : takes
STUDENT ||--o{ STUDENT_GOAL : sets

COURSE_PROVIDER ||--o{ PROVIDER_COURSE : offers
PROVIDER_COURSE ||--o{ ENROLLMENT : referenced_by
ENROLLMENT ||--o{ LMS_EVENT : generates

REQUIREMENT_SET ||--o{ REQUIREMENT : contains
REQUIREMENT ||--o{ REQUIREMENT_MAPPING : maps
REQUIREMENT_MAPPING }o--|| PROVIDER_COURSE : satisfied_by

```

```
PATHWAY ||--o{ PATHWAY_COURSE : includes
PATHWAY_COURSE }o--|| PROVIDER_COURSE : references
```

```
STAFF ||--o{ SUPPORT_TICKET : assigned
AI_SESSION ||--o{ AI_MESSAGE : contains
AI_SESSION ||--o{ AI_AUDIT_LOG : audited
```

```
FRANCHISE {
  uuid id PK
  string legal_name
  string country
  string status
  datetime created_at
}
```

```
CENTER {
  uuid id PK
  uuid franchise_id FK
  string name
  string city
  string country
  string status
  datetime created_at
}
```

```
STAFF {
  uuid id PK
  uuid center_id FK
  string role
  string email
  string status
  datetime created_at
}
```

```
STUDENT {
  uuid id PK
  uuid center_id FK
  string first_name
  string last_name
  date birthdate
```

```
    string preferred_language
    string status
    datetime created_at
}

COURSE_PROVIDER {
    uuid id PK
    string code "ASU/BYU/EXT"
    string name
}

PROVIDER_COURSE {
    uuid id PK
    uuid provider_id FK
    string provider_course_id
    string title
    string credit_value
    string category
}

ENROLLMENT {
    uuid id PK
    uuid student_id FK
    uuid provider_course_id FK
    string term
    string status
    string deep_link_url
    datetime created_at
}

LMS_EVENT {
    uuid id PK
    uuid enrollment_id FK
    string event_type
    json payload
    datetime event_ts
}

REQUIREMENT_SET {
    uuid id PK
    string code "HS_REQ/GE_REQ"
```

```
    string name
    string version
}

REQUIREMENT {
    uuid id PK
    uuid requirement_set_id FK
    string code
    string title
    float units_required
}

REQUIREMENT_MAPPING {
    uuid id PK
    uuid requirement_id FK
    uuid provider_course_id FK
    string rule_type "direct/conditional"
    json rule_params
}

PATHWAY {
    uuid id PK
    string name
    string cluster
    string description
}

PATHWAY_COURSE {
    uuid id PK
    uuid pathway_id FK
    uuid provider_course_id FK
    string priority "must/should/can"
}

STUDENT_ASSESSMENT {
    uuid id PK
    uuid student_id FK
    string type "career/diagnostic"
    json results
    datetime created_at
}
```

```
STUDENT_GOAL {  
  uuid id PK  
  uuid student_id FK  
  string goal_type  
  string goal_value  
  datetime created_at  
}
```

```
AI_SESSION {  
  uuid id PK  
  uuid student_id FK  
  string type "tutor/guidance/writing"  
  string status  
  datetime created_at  
}
```

```
AI_MESSAGE {  
  uuid id PK  
  uuid ai_session_id FK  
  string actor "user/assistant/system"  
  text content  
  datetime created_at  
}
```

```
AI_AUDIT_LOG {  
  uuid id PK  
  uuid ai_session_id FK  
  string policy_version  
  json guardrail_results  
  string risk_level  
  datetime created_at  
}
```

```
SUPPORT_TICKET {  
  uuid id PK  
  uuid student_id FK  
  uuid assigned_staff_id FK  
  string category  
  string priority  
  string status
```

```
text summary
datetime created_at
}
```

4) Security threat model (STRIDE + controls)

4.1 Scope & assets

Assets

- Student PII (LGPD/GDPR-sensitive)
- Academic progress metadata (FERPA-aligned)
- Franchise operational data (financial KPIs)
- AI session content + audit logs
- Integration credentials (LMS deep links, tokens)

Trust boundaries

- Client ↔ API gateway
- Platform ↔ AI provider
- Platform ↔ external universities
- Center networks ↔ cloud services

4.2 STRIDE threats & mitigations

S — Spoofing identity

- **Threats:** account takeover, impersonating center staff, fake LMS webhook sender
- **Controls:**

- MFA (mandatory for staff/admin)
- OIDC + short-lived JWT + refresh rotation
- Signed webhooks (HMAC + IP allowlist)
- Device/session anomaly detection (impossible travel)

T — Tampering

- **Threats:** modify credit mappings, alter progress status, change deep-link URLs
- **Controls:**
 - RBAC/ABAC on write operations
 - Immutable audit trail for: mappings, enrollments, status changes
 - WORM storage for audit logs (or append-only tables)
 - Config-as-code + approvals for mapping rules

R — Repudiation

- **Threats:** staff disputes actions (“I didn’t change it”), AI outputs questioned
- **Controls:**
 - Structured audit events: actor, time, before/after
 - AI session logs + policy version stamping
 - Ticket/CRM correlation IDs for every escalation

I — Information disclosure

- **Threats:** data leaks across franchises, PII exposure in logs, AI prompt leakage
- **Controls:**
 - Tenant isolation at data layer (row-level security)
 - PII encryption at rest; field-level for sensitive

- Secrets manager; never log tokens or PII
- AI prompt redaction + PII minimization
- Access reviews; least privilege by default

D — Denial of service

- **Threats:** API flooding, AI endpoint exhaustion, webhook storms
- **Controls:**
 - WAF + rate limiting per user/tenant
 - Queue-based ingestion (webhooks → queue)
 - Circuit breakers for AI provider calls
 - Autoscaling + backpressure

E — Elevation of privilege

- **Threats:** staff accessing other centers, franchise owners accessing student chat content
- **Controls:**
 - Fine-grained scopes (center_id bound)
 - Just-in-time access for sensitive operations
 - Admin actions require step-up auth
 - Security tests for authorization (ABAC unit tests)

4.3 Security baseline checklist

- OWASP ASVS Level 2 (minimum)
- OWASP Top 10 testing in CI
- SAST/DAST + dependency scanning

- Regular pen test pre-launch and annually
 - Incident response runbook + drills
-

5) AI governance policy (operational + technical)

This is designed to protect:

- academic integrity
- legal posture (non-instructional)
- student safety
- brand trust
- partner university relationships

5.1 AI purpose & scope

AI may:

- guide planning and study habits
- explain concepts at a high level
- provide writing feedback and structure
- recommend pathways and electives
- detect risk and trigger human escalation

AI must not:

- generate answers for graded submissions
- produce plagiarism

- impersonate university faculty
- claim credits, diplomas, or admissions guarantees
- give legal/medical advice beyond safe general info

5.2 Policy enforcement layers

1. **Pre-processing:** PII redaction, prompt sanitation
2. **Policy engine:** allow/deny per session type
3. **Content filters:** plagiarism/cheating detection heuristics
4. **Post-processing:** citations/traceability for advice; disclaimers
5. **Escalation:** auto-ticket for high-risk topics or repeated failure

5.3 Human-in-the-loop rules

Mandatory escalation when:

- student requests “do it for me” on assignments
- mental health self-harm signals
- harassment/abuse signals
- repeated failure in same concept
- compliance-sensitive claims (credits, admissions, visa)

5.4 AI auditability

Every AI session stores:

- policy_version
- risk_level
- categories triggered

- model + parameters (temperature class)
- anonymized prompt hash
- escalation status

Retention:

- default 12 months (configurable)
- deletion/export workflows for LGPD/GDPR

5.5 Model governance

- Approved model list + change control
 - Regression tests on:
 - disallowed behaviors (cheating)
 - factuality templates
 - tone and safety boundaries
 - Quarterly policy review with legal + academics
-

6) Cost & infra sizing model (with assumptions)

This is a **planning model**. Actual costs depend on vendor choices, AI usage, and integration depth.

6.1 Assumptions (tune these)

- MVP Year 1 active students: **2,000**
- Centers: **10**

- DAU/MAU: 25%/100% (500 daily active)
- Average AI sessions per student per week: **3**
- Avg tokens per session (in+out): **2,000**
- Monthly tickets per 100 students: **15**
- LMS event volume: ~20 events/student/month

6.2 Workload estimates

- AI sessions/month: 2,000 students $\times$ 3/wk $\times$ 4 = **24,000**
- Tokens/month: 24,000 $\times$ 2,000 = **48M tokens/month**
- LMS events/month: 2,000 $\times$ 20 = **40,000**

6.3 Infra components and rough monthly cost bands (MVP)

Compute (API + services)

- 3–6 app servers (autoscaled) + background workers
\$800–\$2,500/mo

Database (Postgres managed)

- Primary + read replica + backups
\$400–\$1,500/mo

Cache (Redis)

\$100–\$400/mo

Object storage + CDN

\$50–\$300/mo

Observability (logs/metrics/traces)

\$200–\$1,200/mo (depends on log volume)

Messaging (email/SMS/WhatsApp)

\$200–\$2,000/mo (depends on campaigns)

Data warehouse/BI

\$200–\$1,500/mo

Vector DB

\$200–\$1,500/mo (or self-host cheaper but more ops)

AI spend (biggest variable)

Using the estimate above (48M tokens/mo):

- Depending on model tier and rates: **\$2,000 – \$20,000+/mo**

Total MVP range (excluding heavy AI): ~\$2,150 – \$10,900/mo

With AI: ~\$4,000 – \$30,000+/mo

6.4 Scale model (10× students)

At **20,000 active students**:

- App + DB: ~3–5× (not 10× due to scaling efficiency)
- AI: ~10× unless you reduce tokens per session via:
 - smaller models for most interactions
 - caching answers
 - tighter RAG
 - shorter context windows
 - “coach mode” prompts

6.5 Cost control levers (recommended)

- **Model routing**: cheap model default, premium for hard cases
- **Token budgeting** per student/week

- **Response caching** for repeated FAQs
 - **RAG hygiene** to avoid dumping long context
 - **Batch ingestion** for LMS events and analytics
-

What you want next (so I can lock this into “engineering-ready”)

I can deliver these as **actual files** (recommended for your team):

- C4 diagrams as `.md` + `.mmd`
- OpenAPI as `openapi.yaml`
- ERD as `.mmd`
- Threat model as `.md` (STRIDE table format)
- AI governance policy as `.md` + checklist
- Cost model as `.xlsx` (inputs tab + outputs tab)